

```
1  LIBRARY ieee;
2  USE ieee.std_logic_1164.ALL;
3  USE ieee.numeric_std.ALL;
4  USE work.ALL;
5
6  ENTITY code_lock_simple IS
7      PORT (
8          clk : IN STD_LOGIC;
9          reset : IN STD_LOGIC;
10         code : IN STD_LOGIC_VECTOR(3 DOWNTO 0);
11         enter : IN STD_LOGIC;
12         err_count : OUT STD_LOGIC_VECTOR(1 DOWNTO 0);
13         lock : OUT STD_LOGIC;
14         lock0 : OUT STD_LOGIC
15     );
16 END code_lock_simple;
17
18 ARCHITECTURE rtl OF code_lock_simple IS
19     -----Interne signaler-----
20     SIGNAL enter_sync : STD_LOGIC;
21     SIGNAL failed : STD_LOGIC;
22     SIGNAL err_event : STD_LOGIC;
23
24 BEGIN
25     synchronizer : ENTITY work.synch PORT MAP (clk => clk, async_sig => NOT enter,
26     rise => OPEN, fall => enter_sync); --Brug enten rise eller fall output
27
28     fsm : ENTITY work.code_lock_simple_fsm PORT MAP (clk => clk, reset => reset, enter
29     => enter_sync,
30     code => code, lock => lock, err_event => err_event, failed => failed, lock0 =>
31     lock0);
32
33     wrong : ENTITY work.wrong PORT MAP(clk => clk, reset => reset, failed => failed,
34     err_event => err_event, err_count => err_count);
35 END;
```